

5.1 Exploring Side-Side-Side Triangle Congruence

Lesson Introduction

 Benchmarks	MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg. MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates. MA.912.GR.5.1 Construct a copy of a segment or an angle.		 Learning Target	I can use constructions and transformations to determine if SSS congruence can be used to show that two triangles are congruent.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	- How can I use transformations to determine if SSS congruence can be used to show that two triangles are congruent? - How can I use constructions to determine if SSS congruence can be used to show that two triangles are congruent?			
 Lesson Narrative	In this lesson, students will explore the side lengths of triangles to discover the Side-Side-Side (SSS) congruence postulate. They will use this information to determine if a congruent triangle can be constructed using given side measurements.			
 Component Summary	5.1.1 Warm-Up (~5 min.)	Energy distribution using triangles activity (<i>SE p. 223</i>)	5.1.3 Bring It Together (10 min.)	Summarize that the Side-Side-Side (SSS) Congruence Postulate is true (<i>SE p. 228</i>)
	5.1.2 Exploration (20-25 min.)	Exploration with constructing triangles whose sides are congruent (<i>SE p. 224</i>)	5.1.4 Wrap-Up (~5 min.)	Explanation of whether two triangles are congruent given specific side lengths (<i>SE p. 229</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Side-Side-Side (SSS) Congruence Postulate <u>Additional Terms:</u> - Angles - Congruence - Rigid motions - Transformation		- Compass - Patty Paper - Ruler - Protractor (Optional) Highlighters (Optional) Triangle Strength Video	
	Additional Preparation This lesson is an in-depth exploration. The design of the lesson is not to give students the information, but for them to discover the SSS Congruence Postulate. Consider doing 5.1.2 Exploration on your own to see how it unfolds to make it easier to connect with students.			

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may make errors when using the tools for this lesson (compass, patty paper, ruler, and protractor). - Students may overthink creating a second triangle with the same lengths. - Students may not understand how angles are created. - Students may misread side lengths using line segment notation.	The vast majority of this lesson is 5.1.2 Exploration. Students will be using tools such as a compass and patty paper explicitly during the activity. Consider doing a brief review of the tools prior to launching the lesson if students need that level of support.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Take Turns During 5.1.2 Exploration, students will be working together with a partner or small group to determine if two triangles are congruent using side lengths. Some questions like question 3c, ask students to take turns working with different classmates to share their conjectures for creating triangles whose sides are congruent, both explaining and defending their summary.	MA.K12.MTR.4.1: Discussion and Reflection In 5.1.2 Exploration, students will be making conjectures based on what they learned through their initial exploration. Based on what they discover, students will create possible arguments that they must share, explain, and defend with other classmates.
 Differentiate Instruction	For the 5.1.2 Exploration, consider grouping students together by their understanding of the use of geometric tools (particularly) and performance on constructions. This can allow for differentiated support for this lesson based on knowledge needed to complete some of the steps. 5.1.2 Exploration has activities that lead into the discovery that two triangles are congruent if the three sides of one are congruent to the three sides of the other triangle through various entry-points. Consider the use of manipulatives such as popsicle sticks, even when they are not expressly stated, to use in 5.1.2 Exploration if students are not making connections. 5.1.4 Wrap-Up provides an opportunity for students to display their level of understanding of the Side-Side-Side (SSS) Congruence Postulate. The student responses can provide data for which students may need additional support.	
Lesson Notes:		

5.1.1 Warm-Up: STEM

Component Overview: Students will explore energy distribution using triangles. For this activity, students work independently. There is a video that is available to extend the concept of triangle strength.



5.1.1 Warm-Up: STEM

Triangles are effective tools for architecture used in the design of buildings and other structures, as they provide strength and stability.

When building materials are used to form a triangle, the design has a heavy base and the pinnacle on the top is capable of handling weight because of how the energy is distributed throughout the triangle. The triangle's use in architecture dates back longer than other common architectural shapes such as the dome, arch, and cylinder. It even predates the wheel.

1. Imagine using straws or toothpicks to make a square or rectangle. What do you think will happen when the engineer puts pressure on the side of the rectangle?

Answers will vary. When the engineer puts pressure on the side of the rectangle, the rectangle will fall to the side. The rectangle will break.

2. Now imagine you made a triangle using straws or toothpicks. What do you think will happen when the engineer puts pressure on the side of the triangle?

Answers will vary. When the engineer puts pressure on the side of the triangle, the legs of the triangle will not move. The engineer will have to push harder to break the triangle.



Consider beginning 5.1.1 Warm-Up by reading the narrative language for the whole class to gather buy-in and engagement with the students.

This activity also has a corresponding video that can help engage students in the content of 5.1.1 Warm-Up.



Provide an auditory entry-point by modifying the 5.1.1 Warm-Up activity to a discussion with small groups or the whole class. Questions 1 and 2 allow students to make a guess on what they think will happen. If you do this as a class discussion, it can allow for more ideas to be generated.

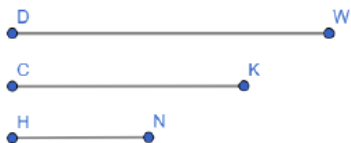
5.1.2 Exploration: Are Three Sides Enough?

Component Overview: Students will explore constructing triangles whose sides are congruent. For this activity, consider having each partner pair or small group work on one question at a time before bringing the class back together to discuss, to ensure that students are on the right track prior to progressing through the next part of the activity.



5.1.2 Exploration: Are Three Sides Enough?

1. $\overline{DW}$, $\overline{CK}$, and $\overline{HN}$ are given. Use the given steps to construct $\triangle BYT$ so that its sides are congruent to $\overline{DW}$, $\overline{CK}$, and $\overline{HN}$.



Step 1: Copy $\overline{DW}$. Construct $\overline{BY}$ so that it is congruent to $\overline{DW}$.

Step 2: Copy $\overline{CK}$. Place the compass on point B to create an arc that is the length of $\overline{CK}$.

Step 3: Copy $\overline{HN}$. Place the compass on point Y to create an arc that is the length of $\overline{HN}$.

Step 4: Mark the intersection of the two arcs using a point. Label the point T .

Step 5: Draw $\overline{BT}$ and $\overline{YT}$.

Answers will vary.



Students may not recall the process for creating a line segment. As a refresher, students can refer to Unit 1, Lessons 8 and 9 on copying a line segment.



The final image for question 1 is a triangle. Answers may vary with some reflection or rotation of the answer shown.

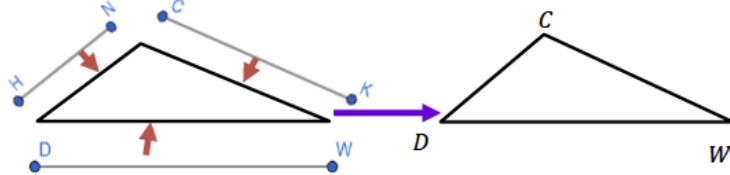


Notice and Wonder

- What do you notice about your resulting triangle?
- What do you wonder about it?

5.1.2 Exploration: Are Three Sides Enough? (continued)

2. The triangle below was created using the three given line segments.



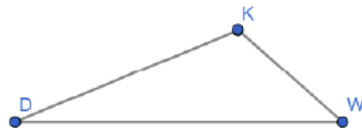
Compare the triangle you constructed with $\triangle CDW$. Describe any similarities or differences.

Answers will vary. $\triangle CDW$ was made using the same lines used to make $\triangle BYT$, so it has the same dimensions. Both triangles have sides that are congruent. The interior angle measures are the same. The triangles are located in different places on the plane and are named differently.

3. Maria used a computer program to create line segments $\overline{DW}$, $\overline{CK}$, and $\overline{HN}$. She then moved the line segments to construct $\triangle DKW$. Because the vertices of the line segments $\overline{CK}$ and $\overline{HN}$ coincided with other vertices, she only used the letters D , K , and W .

- a. What do you notice about $\triangle BYT$ and $\triangle DKW$?

Answers will vary. $\triangle BYT$ looks the same as $\triangle DKW$. $\triangle BYT$ has the same measurements as $\triangle DKW$ because it was made with the same lines.



Discussion prompts for question 2:

- How can you determine that both triangles have the same dimensions?
- How do you know if the triangles have sides that are congruent to one another?



What are the benefits of using a computer program to create line segments? How could you confirm the accuracy of the line segments made from that method?



For question 3a, allow different students to share out what they noticed to check for similarities and potential misconceptions that need to be addressed.

5.1.2 Exploration: Are Three Sides Enough? (continued)

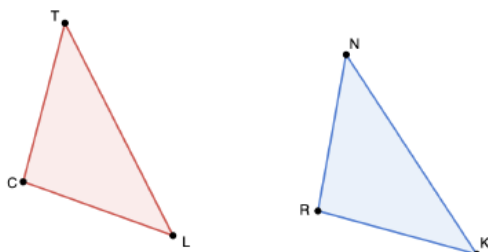
- b. The line segments $\overline{DW}$, $\overline{CK}$, and $\overline{HN}$ were used to construct three different triangles. Work with your partner to write a conjecture about creating triangles whose sides are congruent.

Answers will vary. If a triangle is constructed using the same side lengths, the new triangle will be congruent to the original.

- c. Ask two classmates about their conjectures. Record their conjectures and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conjecture for Creating Triangles Whose Sides are Congruent	My Summary of Their Reason	Initials
<i>Answers will vary.</i>		
<i>Answers will vary.</i>		

4. Use patty paper to determine if $\triangle CTL$ and $\triangle RNK$ were created using the same three line segments.



Answers will vary. Yes, $\triangle CTL$ and $\triangle RNK$ line up perfectly.



For question 3b, listen carefully for students making conjectures that aren't rooted in evidence. For example, some students may make conjectures that are based on their opinion but not based on what they have observed in 5.1.2 Exploration.



For question 3c, consider reviewing the instructions with the students to have a better idea of what is expected from this activity.



For a challenge, have students find one additional student who has a conjecture they agree with and one student who has a conjecture they don't agree with. For both cases, have students explain why they do or don't agree with that conjecture.



Students may need additional support when it comes to the use of patty paper. Consider providing a brief tutorial for students on the use of patty paper prior to releasing them to work on question 4.

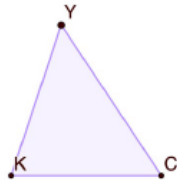
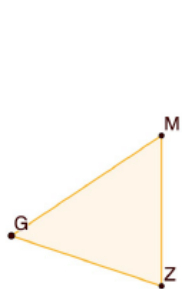
5.1.2 Exploration: Are Three Sides Enough? (continued)

5. Discuss the following with your partner: If three sides of a triangle are congruent, does that mean all of the angles of the triangles are congruent too? Explain your decision and state the evidence used for the decision.

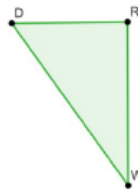
Answers will vary. Yes, all angles are congruent. If three sides of a triangle are congruent, then all the angles of the triangles are congruent because each pair of sides will create the same angles in order to fit together to make a triangle.

6. Two sets of triangles are shown.

$\triangle MGZ$ and $\triangle KYC$



$\triangle HTX$ and $\triangle WDR$



- Use patty paper to determine if the sides of the two triangles are congruent.
- $\triangle MGZ$ and $\triangle KYC$ are congruent because all of the sides are congruent.
- For the two triangles that are congruent, use patty paper to verify that the angles are congruent.
- For the two triangles that are congruent, list the sides on each triangle that are congruent to each other.

$\overline{ZG} \cong \overline{KY}$, $\overline{CK} \cong \overline{MZ}$, and $\overline{CY} \cong \overline{GM}$



Notice and Wonder

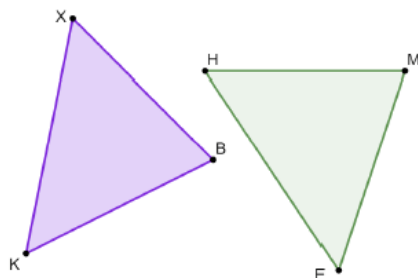
- What do you notice about the two sets of triangles presented in question 6?
- What are some things that you wonder about these triangles?



When reviewing with the class, consider allowing a student to come to the board or document camera to show how they used the patty paper to determine that corresponding sides and angles of the two triangles were congruent.

5.1.2 Exploration: Are Three Sides Enough? (continued)

7. $\triangle KXB$ and $\triangle HFM$ are shown. $HM = 4.2$, $XB = 4.2$, $XK = 5$, and $KB = 4.4$.



- a. Discuss with your partner if the given measurements are sufficient to determine whether $\triangle KXB$ and $\triangle HFM$ are congruent. Explain your decision.

Answers will vary. The given measurements are not sufficient to determine if the triangles are congruent because we do not know if all sides are the same length. We only know that $\overline{XB} \cong \overline{HM}$.

- b. Use patty paper, a protractor, a ruler, or a compass to measure all of the parts of $\triangle KXB$ and $\triangle HFM$. Then, complete the sentence.

$\triangle KXB$ and $\triangle HFM$ are
☐ congruent
☐ not congruent
 because
☐ 0
☐ 1
☐ 2
☒ 3

 side(s) and
☐ 0
☐ 1
☐ 2
☒ 3
 angle(s) are congruent.



Consider using colored highlighters or markers to illustrate the connection between the side lengths that are congruent to one another.



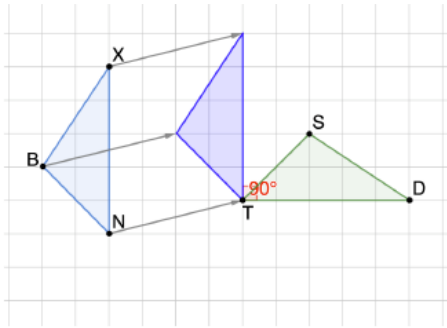
What process did you and your partner go through to decide whether the given measurements are sufficient for determining if the triangles are congruent?



- What tool(s) did you use to complete the statement in question 7b?
- Why did you choose that tool?

8. A series of transformations shows how $\overline{BX}$, $\overline{XN}$, and $\overline{BN}$ were translated, then rotated 90° about point T to create $\overline{SD}$, $\overline{DT}$, and $\overline{TS}$.

5.1.2 Exploration: Are Three Sides Enough? (continued)



- a. Use the grid to determine if $BN = TS$, $BX = DS$, and $XN = TD$.

Answers will vary. Yes, all corresponding sides of the triangle are congruent.

- b. Discuss with your partner how the transformations that were used ensure that the length of each line segment was preserved.

Answers will vary. The transformations that were used only changed the location and not the length of each side of the triangle. Therefore, the transformations were rigid transformations and preserved the lengths of all sides of the triangle.

- c. Use the word bank to complete the statements.
Words may be used more than once.

Word Bank

angles	congruent	congruence	parts	rigid motions
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Since $\overline{BX}$, $\overline{XN}$, and $\overline{BN}$ were translated then rotated 90° , the length of each line segment was preserved because of the definition of congruence in terms of rigid motions. The definition of congruence in terms of rigid motions also ensures that the angles are congruent. Because all of the parts of $\triangle XNB$ are congruent to all of the corresponding parts of $\triangle DST$, the two triangles are congruent.



Prompts to consider for the image of question 8:

- What do you notice about the images?
- What do you wonder about the images?
- What do you know about translations?
- What do you know about rotations?



In question 8b, students may not recall the term “preserved” from Unit 1 Lesson 2 and may need additional support. Consider defining what this word means within the context prior to releasing students to work on the question.

For question 8b, students are not required to write down their answers. They will use the discussion from question 8b to complete the statement in question 8c. For question 8c, let students know that word choices can be used more than once throughout the statement. Consider reviewing question 8c during the 5.1.3 Bring It Together.

5.1.3 Bring It Together: Triangle Congruence

Component Overview: Students will explain, using what they learned in 5.1.2 Exploration, that the Side-Side-Side (SSS) Congruence Postulate is true. For this activity, students should work with their partner to complete the questions.



5.1.3 Bring It Together: Triangle Congruence



Side-Side-Side (SSS) Congruence Postulate

Two triangles are congruent if the three sides of one are congruent to the three sides of the other, respectively.

1. Work with your partner to explain how the exploration has shown that the Side-Side-Side (SSS) Congruence Postulate is true.

Answers will vary. The explorations showed us Side-Side-Side is true by using three different segments to create congruent triangles. The triangles with three congruent sides also had congruent angles, so three congruent sides were enough to show the triangles were congruent.



Students may need a refresher on the term “postulate” and how it applies to Side-Side-Side.



After giving students time to discuss, consider asking students to share with the class what they discussed with their partner. Also, allow other students to critique their reasoning and provide feedback.

5.1.4 Wrap-Up: Cool Down

Component Overview: Students will explain whether two triangles are congruent given specific side lengths. Students should complete this activity independently so that accurate student-level data on their understanding of the lesson content can be collected.



5.1.4 Wrap-Up: Cool Down

1. Maggie and Naomi both constructed a triangle using the side lengths shown.

- $CN = 2.5$ inches (in.)
- $NK = 3.75$ in.
- $CK = 5.25$ in.

Explain whether it is possible to find out if the two triangles Maggie and Naomi created will be congruent.

Answers will vary. Since Maggie and Naomi created triangles from the same three side lengths, their triangles will be congruent by SSS Congruence Postulate.



This question assesses student understanding of the Side-Side-Side (SSS) Congruence Postulate. The data from the student response can provide insight on whether or not the students need additional support on the content from this lesson.

When working with students in small groups who struggled with the Ticket Out the Door, provide students with paper strips with three lengths and ask them to create a triangle. Then, allow each member of the small group to share what they notice and wonder about the triangles.